MIL-T-52502B
25 September 1980
SUPERSEDING
MIL-T-52502A
23 October 1975

MILITARY SPECIFICATION

TRAINING AID, PANELS, DECLINATION DIAGRAM AND GRAPHIC SCALE, MAP READING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of panels.

1.2 Classification. Panels covered by this specification shall be of the following types as specified (see 6.2):

Type I - Declination
Type II - Graphic scale

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6910

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
UU-P-268	- Paper, Kraft, Wrapping.
LLL-B-810	- Building Board (Hardboard), Hard Pressed, Vegetable Fiber.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking For Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The training aids shall consist of panels of building board with declination diagram and graphic scale markings as shown on Figures 1 and 2, and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

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3.3 Construction.

3.3.1 Building board. Building board shall conform to LLL-B-810. Type II, Surface 2, Finish C, Design a, Size 1/8-inch thick.

3.4 Dimensions.

3.4.1 Type I, declination panel.

3.4.1.1 Panel size. The size of the Type I panel shall be 11.00 inches wide x 17.00 inches high.

3.4.1.2 Compass lines. The magnetic north, grid north, and true north compass lines, with symbols as shown on Figure 1, shall be 0.50 inch wide.

3.4.1.3 Character sizes. Letters shall be gothic, sans serif, 1.25 inches high; numerals shall be gothic, sans serif, 1.75 inches high, and shall be positioned on the panel as shown on Figure 1.

3.4.1.4 Tolerances. Tolerances on all dimensions of Type I panel shall be plus or minus 0.06 inch.

3.4.2 Type II, graphic scale panel.

3.4.2.1 Panel size. The size of the Type II panel shall be 50.00 inches wide x 7.00 inches high.

3.4.2.2 Scale size. The scale shall be 42.00 inches wide x 1.50 inches high. The scale shall be positioned 0.50 inch above the bottom edge of the panel.

3.4.2.3 Character sizes. The numerals representing distances in meters and the abbreviation "m" shall be gothic, sans serif, 1.250 inches high. The scale ratio letters and numerals shall be gothic, sans serif, 1.625 inches high. The characters shall be positioned on the panel as shown on Figure 2.

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3.4.2.4 Line sizes. The scale border lines and vertical meter division lines shall be 0.125 inch thick and the horizontal meter length lines shall be 0.50 inch thick as shown on Figure 2.

3.4.2.5 Tolerances. Tolerances on all dimensions of Type II panel shall be plus or minus 0.06 inch.

3.5 Treatment and painting. The portions of the training aid panels normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. Background surface shall be painted white. Lines and characters shall be painted black.

3.6 Workmanship. Particular attention shall be given to neatness and legibility of all markings. Panels shall have smooth edges free from spots, blisters, tears, scuffs, and checks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The panels shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Dimensions not as specified.
103. Marking missing, incomplete, or illegible.
104. Treatment and painting not as specified.
105. Workmanship not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Materials, methods, or containers not as specified for Level A or B. Each incorrect material, method, or container shall constitute one defect.
107. Panels bent.
108. Consolidation not as specified for Level A or Level B.
109. Gross weight or size exceeds the limitation of the box specification for Level A or Level B.
110. Strapping not as specified for Level A.
111. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B or Commercial as specified (see 6.2).

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5.1.1 Level A. Each panel shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3. Bending of the panel shall not be permitted.

5.1.1.1 Consolidation. The preserved panels of like description shall be consolidated together in a close-fitting box conforming to PPP-B-636, W5c, style optional, in quantities not to exceed the weight limitation of the box specification.

5.1.2 Level B. Each panel shall be wrapped with paper conforming to UU-P-268, Type I, grade optional, and the wrap secured with tape conforming to PPP-T-45, Type III, Grade A. Bending of the panel shall not be permitted.

5.1.2.1 Consolidation. The preserved panels of like description shall be consolidated together in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 175, style optional, in quantities not to exceed the weight limitation of the box specification.

5.1.3 Commercial. Panels shall be preserved in accordance with MIL-STD-1188. Bending of the panel shall not be permitted.

5.2 Packing. Packing shall be Level A, Level B or Commercial as specified (see 6.2).

5.2.1 Level A. Panels of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification, except that strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable; and, unless otherwise specified herein, shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Panels of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-636, V3c, style optional, or PPP-B-640, Class 2, style optional. The gross weight or size shall not exceed the limitation of the box specification.

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5.2.3 Commercial. Panels, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The declination panel and graphic scale panel are components of a map reading set. The declination panel is used as an aid in teaching the use of the declination diagram printed on military maps and the graphic scale panel is used as an aid in teaching the use of the graphic scale printed on military maps.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When Finish A strapping is required (see 5.2.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodians:
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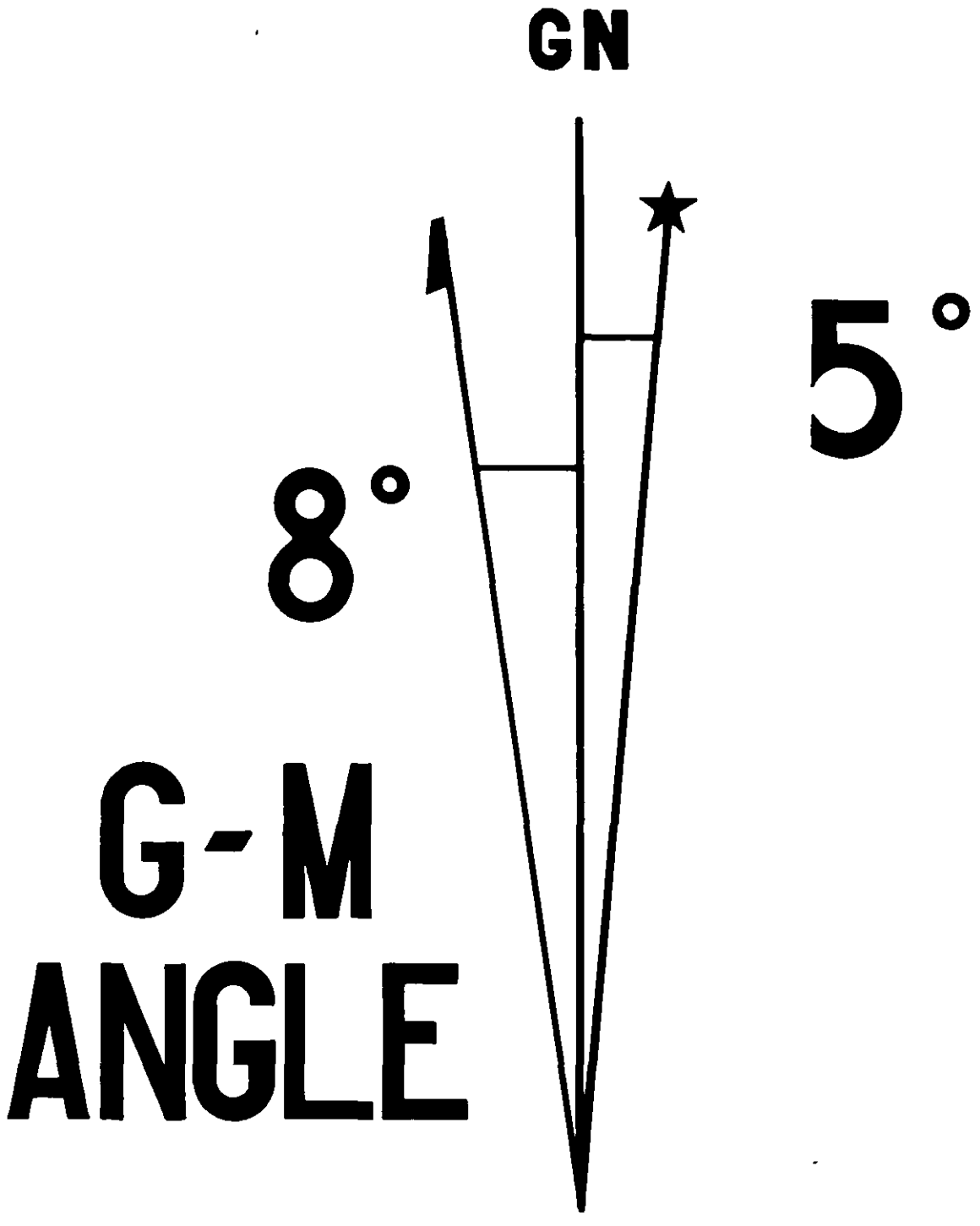


FIGURE 1. Declination diagram.

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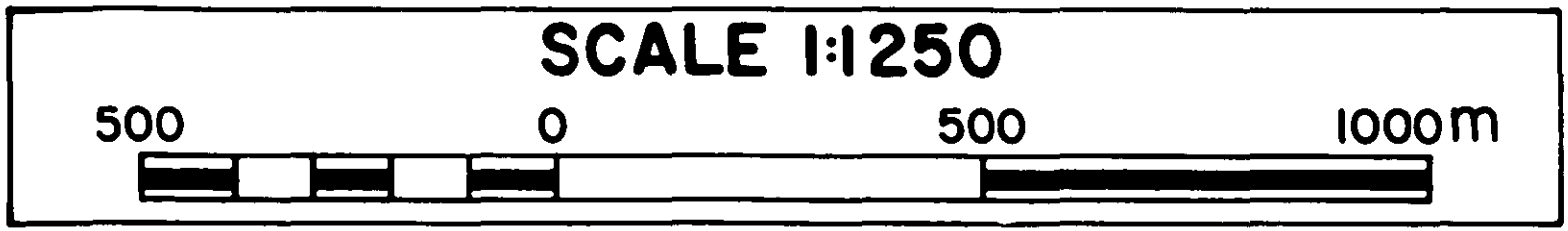


FIGURE 2. Graphic scale.

X-3041

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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